

Error Clustering — Plain-Language Explainer

Real-Time AI Evaluation Engine — Weak Area Detection module

It's the system's way of asking “*is there a pattern in how this student gets things wrong, not just how often?*” — instead of just counting wrong answers, it looks at **how** each wrong answer happened (how fast, how many tries, how hard the question was) and groups similar mistakes together to figure out the **type** of struggle, not just the fact of it.

Every wrong answer gets tagged with 4 details — skill, answer time, number of attempts, and question difficulty — and a clustering algorithm (KMeans) groups the student's recent wrong answers into one of five behaviour types:

Pattern	What it looks like
Impulsive	Answering wrong, fast — guessing without really reading
Systematic	Same type of mistake, repeatedly — a genuine misconception
Struggling	Slow <i>and</i> wrong, repeatedly, on the same skill — a real knowledge gap
Difficulty spike	Only wrong on the hardest questions — actually knows the basics fine
Careless	One-off wrong answer, inconsistent with otherwise-good performance — just a slip

The point: two students can both be “at 40% accuracy,” but one is guessing carelessly and the other has a real gap — this tells the tutor *which one* it's dealing with, so it can react differently (slow down and ask them to read carefully vs. actually re-teach the concept).

Source: [personalized-robot- / code / Backend / app / ml / error_clustering.py](#)